Assignment 1

Convex Optimisation

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Question 1

Let $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$, where the system $Ax = b$ is consistent and admits many solutions.

a) (ℓ_0)-sparse solution

Maximizing sparsity is equivalent to solving the following constrained optimization problem:

$$\min_{x\in\mathbb{R}^n} \|x\|_0 \quad \text{s.t. } Ax = b \tag{1}$$

For this to be a convex optimization problem:

1. The objective function to minimise should be convex.
2. The constraint set must be convex.

Checking whether the objective function is convex :

$$\text{obj}(x) = \|x\|_0, \quad x \in \mathbb{R}^n \tag{2}$$

Consider two points $x_1, x_2 \in \mathbb{R}^n$. We define $x_3(\lambda)$ as the convex combination of these two points.

$$x_3(\lambda) = \lambda x_1 + (1-\lambda)x_2, \quad \lambda \in [0,1] \tag{3}$$

For the chosen objective function, we have

$$\lambda \,\text{obj}(x_1) + (1-\lambda)\,\text{obj}(x_2) = \lambda\|x_1\|_0 + (1-\lambda)\|x_2\|_0, \tag{4}$$

$$\text{where } \lambda \in [0,1], \quad \lambda\|x_1\|_0 + (1-\lambda)\|x_2\|_0 \in [\min\{\|x_1\|_0, \|x_2\|_0\}, \max\{\|x_1\|_0, \|x_2\|_0\}]\,. \tag{5}$$

It is possible to choose $x_1, x_2 \in \mathbb{R}^n$ and $\lambda \in [0,1]$ such that

$$\max\{\|x_1\|_0, \|x_2\|_0\} < n, \tag{6}$$

$$\|\lambda x_1 + (1-\lambda)x_2\|_0 = n. \tag{7}$$

Which clearly tells that obj(x) is not a convex function, since this would mean that there exists $x_1, x_2 \in \mathbb{R}^n$ and $\lambda \in [0,1]$ such that

$$\text{obj}\left(\lambda x_1 + (1-\lambda)x_2\right) > \lambda\,\text{obj}(x_1) + (1-\lambda)\,\text{obj}(x_2)$$

Counter example where obj(x) violates the rule for convexity :
Consider $x_1 \in \mathbb{R}^n$. Define

$$x_1 = \begin{bmatrix} a_1 & a_2 & \dots & a_\alpha & 0 & 0 & \dots & 0 \end{bmatrix}^T, \quad a_i > 0,\ i \in \{1, 2, \dots, \alpha\},$$

so that

$$\|x_1\|_0 = \alpha < n.$$

Similarly, let

$$x_2 = \begin{bmatrix} 0 & 0 & \dots & 0 & b_{\alpha+1} & b_{\alpha+2} & \dots & b_n \end{bmatrix}^T, \quad b_i > 0,\ i \in \{\alpha+1, \alpha+2, \dots, n\},$$

so that

$$\|x_2\|_0 = n - \alpha < n.$$

Choosing $\lambda = \frac{1}{2}$, we obtain

$$\text{obj}(\lambda x_1 + (1-\lambda)x_2) = n,$$

since

$$\lambda x_1 + (1-\lambda)x_2 = \begin{bmatrix} a_1/2 & a_2/2 & \dots & a_\alpha/2 & b_{\alpha+1}/2 & \dots & b_n/2 \end{bmatrix}^T$$

while

$$\lambda \operatorname{obj}(x_1) + (1-\lambda)\operatorname{obj}(x_2) = \tfrac{n}{2}.$$

Hence, for $\lambda = \frac{1}{2}$,

$$\operatorname{obj}(\lambda x_1 + (1-\lambda)x_2) \;>\; \lambda \operatorname{obj}(x_1) + (1-\lambda)\operatorname{obj}(x_2).$$

which proves that the objective function to minimise is not convex and hence, the optimisation problem is not convex.

b) (ℓ_1)-proxy The ℓ_1-relaxation of the ℓ_0-sparse solution problem is given by

$$\min_{x\in\mathbb{R}^n} \|x\|_1 \quad \text{s.t. } Ax = b, \tag{8}$$

where we define the objective function to minimize as

$$\operatorname{obj}_2(x) = \|x\|_1. \tag{9}$$

This is a convex optimization problem because:

1. The objective function $\operatorname{obj}_2(x)$ is convex.
2. The constraint set $\{x \in \mathbb{R}^n : Ax = b\}$ is convex.

Proof.

1. **Convexity of the objective function.** The objective function to minimize, $\operatorname{obj}_2(x)$, is given by

$$\operatorname{obj}_2(x) = \|x\|_1 = \sum_{i=1}^{n} |x_i|. \tag{10}$$

Let $x_1, x_2 \in \mathbb{R}^n$ and let $\lambda \in [0,1]$ be arbitrary. Then

$$\begin{aligned}
\operatorname{obj}_2(\lambda x_1 + (1-\lambda)x_2) &= \|\lambda x_1 + (1-\lambda)x_2\|_1 \\
&= \sum_{i=1}^{n} |\lambda (x_1)_i + (1-\lambda)(x_2)_i| \\
&\le \sum_{i=1}^{n} \Big(|\lambda (x_1)_i| + |(1-\lambda)(x_2)_i| \Big) \\
&= \lambda \sum_{i=1}^{n} |(x_1)_i| + (1-\lambda) \sum_{i=1}^{n} |(x_2)_i| \\
&= \lambda \operatorname{obj}_2(x_1) + (1-\lambda)\operatorname{obj}_2(x_2),
\end{aligned}$$

where we used the inequality $|a+b| \le |a| + |b|$ for all $a, b \in \mathbb{R}$.

Hence, $\operatorname{obj}_2(x)$ is convex.

2. **Convexity of the constraint set.**

Let the constraint set be

$$S = \{\, x \in \mathbb{R}^n \;\mid\; Ax = b \,\}.$$

Choose $x_1, x_2 \in S$. Then

$$Ax_1 = b \quad \text{and} \quad Ax_2 = b.$$

Consider the convex combination

$$x_3(\lambda) = \lambda x_1 + (1-\lambda)x_2, \quad \lambda \in [0,1].$$

Applying A, we obtain

$$Ax_3(\lambda) = A(\lambda x_1 + (1-\lambda)x_2) = \lambda A x_1 + (1-\lambda) A x_2 = \lambda b + (1-\lambda) b = b.$$

Thus $x_3(\lambda) \in S$ for all $\lambda \in [0,1]$. Hence, the constraint set S is convex.

Hence, since both the objective function and the constraint set are convex in nature, the optimsation is a convex optimisation problem.

c) Proving Equivalence
To prove the equivalence between the optimisation problem given in this subpart and the one given in equation 30, we must prove both implications:

1. **Forward direction:**
   Assume the problem given in this subpart and transform it into the problem stated in equation 30.
2. **Reverse direction:**
   Assume the problem in equation 30 and transform it into the form given in this subpart.

By completing both steps, we demonstrate that the linear programming formulation is equivalent to the constrained ℓ_1-minimisation problem.

Proving the Forward Direction

Let S_1 denote the set of all $x \in \mathbb{R}^n$ such that

$$\forall x \in S_1, \quad Ax = b \tag{11}$$

Hence, we can view S_1 as

$$S_1 = \{x_1, x_2, x_3, \ldots \mid Ax_i = b,\ i = 1, 2, 3, \ldots\} \tag{12}$$

For a given $x \in S_1$, we can choose $u \in \mathbb{R}^n$ satisfying

$$-u \leq x \leq u, \quad u \geq 0 \tag{13}$$

in many ways.
We need to minimise

$$1^\top u = \sum_{i=1}^{n} u_i \tag{14}$$

with respect to u, for a given x.
For a given x, setting

$$u = |x| \tag{15}$$

i.e.,

$$u_i = |x_i|, \quad i = 1, 2, \ldots, n \tag{16}$$

minimises the quantity in (14) and also satisfies the constraints in (13).
This was about choosing $u \in \mathbb{R}^n$, given a fixed x.

Since the first problem given in the question is

$$\min_{x,\,u} \ \ 1^\top u \quad \text{subject to} \quad Ax = b, \quad -u \leq x \leq u, \quad u \geq 0 \tag{17}$$

where the objective must be minimised with respect to u as well. Then, using equation (15) as the solution of u for a given x, we can reformulate the problem in equation (17) as:

$$\min_{x \in \mathbb{R}^n} \ 1^\top |x| \quad \text{subject to} \quad Ax = b \tag{18}$$

which can be written as

$$\min_{x \in \mathbb{R}^n} \ \|x\|_1 \quad \text{subject to} \quad Ax = b \tag{19}$$

2) Proving The Reverse Direction

In this, we assume the problem

$$\min_{x \in \mathbb{R}^n} \|x\|_1 \quad \text{subject to} \quad Ax = b \tag{20}$$

and prove that it can be recast into

$$\min_{x,\, u \in \mathbb{R}^n} 1^\top u \quad \text{subject to} \quad Ax = b,\ -u \leq x \leq u,\ u \geq 0 \tag{21}$$

We denote the constraint set as S.

Hence, we have

$$\forall\, x \in S, \quad Ax = b \tag{22}$$

We can imagine the set S to be a collection of points $x_i,\ i \in \{1, 2, 3, \ldots\}$, such that

$$S = \{x_1, x_2, x_3, \ldots \mid Ax_i = b,\ i \in \{1, 2, 3, \ldots\}\} \tag{23}$$

For every $x_i \in S$, we define the set S_i as follows:

$$S_i = \{\alpha \in \mathbb{R}^n : \alpha \geq 0,\ -\alpha \leq x \leq \alpha\} \tag{24}$$

(where $\geq$ and $\leq$ for $x, \alpha \in \mathbb{R}^n$ denote elementwise inequalities)

We can say that for a fixed $x_i \in S$,

$$\|x_i\|_1 = \arg\min_{\alpha \in S_i} 1^\top \alpha \tag{25}$$

by just observing the way in which the set S_i is defined.

Since for a given $x_i \in S$, equation (25) holds, and since the problem in equation (20) requires optimisation over all $x \in S$, we can recast it into the problem in equation (21)

Hence, we have proved that the two optimisation problems are equivalent.

d 1)

We are given $A \in \mathbb{R}^{3\times 3}$ where each column of A represents the reflectance of that pigment for R, G, B.
We wish to combine the pigments in a certain way to get the desired R, G, B reflectance $b \in \mathbb{R}^3$.
Let $x \in \mathbb{R}^3$ be the vector representing the way the 3 pigments need to be combined.
We have the following constraint on x:

$$Ax = b \tag{26}$$

The constraint set in equation (26) admits multiple solutions. We wish to find the x which requires the fewest pigments.
This can be formulated as a sparse recovery problem as follows:

$$\min_{x \in \mathbb{R}^3} \|x\|_0 \qquad \text{subject to} \quad Ax = b \tag{27}$$

We now solve the problem in equation (26) manually.

Let the constraint set be S,

$$x \in S, \quad Ax = b$$

Let $x = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix}^T$
For $x \in S$, we need the following to be satisfied:

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0.2 \\ 0.8 \\ 1.0 \end{bmatrix}$$

This system of 3 equations contains only 2 independent equations, which are:

$$x_1 + x_3 = 0.2 \tag{28}$$
$$x_2 + x_3 = 0.8 \tag{29}$$

Hence, to uniquely determine x, we need to fix one of x_1, x_2, x_3. To find the solution x which solves the problem, we use trial and error.

We fix one $x_i = 0$ ($i \in \{1, 2, 3\}$) and solve equations 28 and 29 for x_j ($j \in \{1, 2, 3\},\ j \neq i$).
and see which solution minimises $\|x\|_0$.

1. $x_1 = 0$:
$$x_2 = 0.6,$$
$$x_3 = 0.2$$

2. $x_2 = 0$:
$$x_1 = 0.6,$$
$$x_3 = 0.8$$

3. $x_3 = 0$:
$$x_1 = 0.2,$$
$$x_2 = 0.8$$

All the x in (1), (2), and (3) each have $\|x\|_0 = 2$, and all of them are solutions to the problem in equation (27).

d 2)

We cannot directly solve the problem in equation (27) using CVXPY, as the objective function to minimize $\|x\|_0$ is not convex.

We can instead solve the following problem:

$$\min_{x \in \mathbb{R}^n} \|x\|_1 \quad \text{subject to} \quad Ax = b \tag{30}$$

as the ℓ_1 norm is the closest convex substitute for $\|x\|_0$. It is also the convex envelope of $\|x\|_0$ on the unit ℓ_∞ ball.

$$\text{Unit } \ell_\infty \text{ ball:} \quad \{x \in \mathbb{R}^n : \|x\|_\infty \leq 1\}$$

The optimisation problem in equation (30) often admits solutions close to the solution of the problem in (27).

The paper [1] introduces a more intuitive and a better way to approximate the sparse recovery optimisation problem, such that it can be solved using convex optimisation techniques.

The difference between the ℓ_0 and ℓ_1 objectives as minimisation functions is that larger coefficients are penalised more heavily in ℓ_1 than smaller coefficients, while the ℓ_0 norm "democratises" penalisation.

To address this difference, the paper [1] introduces an iterative algorithm, where in each iteration a weighted formulation of the ℓ_1 norm is minimised. The weights are inversely proportional to the value of the corresponding component of x_i in that iteration, thereby acting as a normalisation factor.

Hence, in each iteration of the algorithm, the objective function is like a convex approximation of the ℓ_0 norm.

The optimizer $x^\star$ given by the algorithm above:

$$x^\star = \begin{bmatrix} 3.0 \times 10^{-23} & 6.00 \times 10^{-1} & 2.0 \times 10^{-1} \end{bmatrix}^T$$

Algorithm 1 proposed in [1]

1. Set iteration count $l = 0$ and $w_i^0 = 1$ for $i = 1, 2, \ldots, n$.
2. Solve the convex optimisation problem of minimising the weighted ℓ_1 norm:
$$x^l = \arg\min_{x \in \mathbb{R}^n} \sum_{i=1}^{n} w_i^l |x_i| \quad \text{subject to} \quad Ax = b \tag{31}$$
3. Update weights for each $i = 1, 2, \ldots, n$:
$$w_i^{l+1} = \frac{1}{|x_i^l| + \varepsilon} \tag{32}$$
where $\varepsilon > 0$ and $\varepsilon \ll 1$ to prevent division by zero.
4. Terminate on convergence or when l attains the maximum number of iterations.

If we assume $3.0 \times 10^{-23} \approx 0$,

$$\text{support}\,(x^\star) = \{2, 3\}$$

e)

This time, A and b are given by

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 0.2 \\ 0.8 \\ 0.95 \end{bmatrix}$$

which makes $Ax = b$ infeasible.

e 1)

We are now in search for $x \in \mathbb{R}^3$ such that

a) $\|Ax - b\|_2$ is minimized. Here, $\|\cdot\|_2$ denotes the ℓ_2 norm of a vector. We want a way to combine the three pigments such that they produce the R, G, B reflectance given by b, but since the system $Ax = b$ is infeasible, we would like a way of combining them to give something close to b.

b) We also want to minimize $\|x\|_0$, since we are in search for the sparse solution.

Hence, we now modify the problem to solve as

$$\min_{x \in \mathbb{R}^3} \lambda \|Ax - b\|_2 + \|x\|_0 \tag{33}$$

where $\lambda > 0$ is a parameter we choose, which will decide the weightage we give to each constraint.

A convex optimisation problem which on solving yields a solution similar to the problem in (30) is given by

$$\min_{x \in \mathbb{R}^3} \lambda \|Ax - b\|_2 + \|x\|_1 \tag{34}$$

However, we choose to solve the problem in the way we did in the previous subpart, which is to do it in iterations and to replace $\|x\|_0$ by a weighted version of $\|x\|_1$ in every iteration.

The additional term $\lambda \|Ax - b\|_2$, being a convex function of x, does not create problems.

A much simpler convex approximation to the problem defined in (33) would be the optimisation problem in (34).

Algorithm 2 Slightly modified version of Algorithm 1

1. Set iteration count $l = 0$ and $w_i^0 = 1$ for $i = 1, 2, \ldots, n$.
2. Solve the following convex optimisation problem:

$$x^l = \arg\min_{x \in \mathbb{R}^3} \lambda \|Ax - b\|_2 + \sum_{i=1}^{n} w_i^l |x_i|$$

3. Update the weights for each $i = 1, 2, \ldots, n$:

$$w_i^{l+1} = \frac{1}{|x_i^l| + \varepsilon}$$

   where $\varepsilon > 0$, $\varepsilon \ll 1$ to prevent division by zero.
4. Terminate on convergence or when l attains the max number of iterations.

e 2)

We solve both, the simplex convex approximation which is 34 as well as the iterative algorithm defined in 2, using CVXPY.

Result:
Taking $\lambda = 10$,

1. Solving the problem in (34) gives the optimizer

$$x^\star = \begin{bmatrix} 3.24 \times 10^{-9} & 5.9 \times 10^{-1} & 1.8 \times 10^{-1} \end{bmatrix}^T.$$

   Assuming $3.24 \times 10^{-9} \approx 0$, the support of $x^\star$ is $\{2, 3\}$. Moreover,

$$\|Ax^\star - b\|_2 = 0.029.$$

2. Solving using the algorithm defined in Algorithm 2 gives the optimizer

$$x^\star = \begin{bmatrix} 3.85 \times 10^{-16} & 6.07 \times 10^{-1} & 1.76 \times 10^{-1} \end{bmatrix}^T.$$

   Assuming $3.85 \times 10^{-16} \approx 0$, the support of $x^\star$ is $\{2, 3\}$. Moreover,

$$\|Ax^\star - b\|_2 = 0.030.$$

Both methods yield similar solutions.

Question 2

We are given:

$$A \in \mathbb{R}^{m \times p}, \quad X \in \mathbb{R}^{p \times n}, \quad B \in \mathbb{R}^{m \times n}$$

We are also told to assume that A is rank deficient or $p > m$ so that the equation

$$AX = B$$

admits multiple solutions.

a) Rank minimization

We are given the constrained optimization problem

$$\min_{X \in \mathbb{R}^{p \times n}} \text{rank}(X) \quad \text{s.t.} \quad AX = B$$

and we need to determine if the problem in equation is convex or not.

Checking if the constraint set is convex:

Let the constraint set be denoted as S. Hence, S is given as

$$S = \left\{ X \in \mathbb{R}^{p \times n} : AX = B \right\}$$

Consider $X_1, X_2 \in S$. Hence, we have

$$AX_1 = B$$
$$AX_2 = B$$

We construct a new point in $\mathbb{R}^{p \times n}$ as follows

$$X_3(\lambda) = \lambda X_1 + (1 - \lambda) X_2, \quad \text{where } \lambda \in [0, 1]$$

Checking if $X_3(\lambda) \in S$ for all $\lambda \in [0, 1]$:

$$AX_3(\lambda) = A(\lambda X_1 + (1 - \lambda) X_2) = \lambda AX_1 + (1 - \lambda) AX_2 = \lambda B + (1 - \lambda) B = B$$

Hence, $X_3(\lambda) \in S$ for all $\lambda \in [0, 1]$.

The constraint set S is convex.

Checking for convexity of the objective function to minimize:

We define the objective function as follows:

$$\text{obj}(X) = \text{rank}(X) \tag{35}$$

where $X \in \mathbb{R}^{p \times n}$.

$\text{obj}(X)$ is not a convex function as:

- $\text{obj}(X) = \text{rank}(X) \leq \min\{p, n\}$
- It is possible to come up with $X_1, X_2 \in \mathbb{R}^{p \times n}$ and $\lambda \in (0, 1)$ such that:
  1. $\text{obj}(X_1) = R_{X_1}, \qquad \text{where } R_{X_1} < \min\{p, n\}$
  2. $\text{obj}(X_2) = R_{X_2}, \qquad \text{where } R_{X_2} < \min\{p, n\}$
  3. $\text{obj}(\lambda X_1 + (1 - \lambda) X_2) = \min\{p, n\}$

     Using 1 and 2 we can say that

     $$\lambda \text{obj}(X_1) + (1 - \lambda) \text{obj}(X_2) \in (\min\{R_{X_1}, R_{X_2}\}, \max\{R_{X_1}, R_{X_2}\})$$

     and therefore,

     $$\lambda \cdot \text{obj}(X_1) + (1 - \lambda) \cdot \text{obj}(X_2) < \min\{p, n\}$$

     which implies

     $$\lambda \cdot \text{obj}(X_1) + (1 - \lambda) \cdot \text{obj}(X_2) < \text{obj}(\lambda X_1 + (1 - \lambda) X_2)$$

A counterexample to show that $\text{obj}(X)$ **is not convex :**

Consider $p = n = 2$. We assume X_1, X_2 to be the following matrices.

$$X_1 = \begin{bmatrix} \frac{1}{2} & \frac{2}{4} \end{bmatrix}$$
$$X_2 = \begin{bmatrix} \frac{3}{7} & \frac{21}{49} \end{bmatrix}$$

Hence, according to our definition of $\text{obj}(X)$ in 35 in we have

$$\text{obj}(X_1) = 1, \qquad \text{obj}(X_2) = 1$$

Let $\lambda = \frac{1}{2}$, and we define $X_3(\lambda)$ as

$$X_3(\lambda) = \lambda X_1 + (1 - \lambda) X_2 \tag{36}$$

Substituting $\lambda = \frac{1}{2}$ in equation 36, we obtain

$$\begin{aligned} X_3\left(\tfrac{1}{2}\right) &= \tfrac{1}{2} X_1 + \tfrac{1}{2} X_2 \\ &= \begin{bmatrix} \frac{4}{2} & \frac{23}{2} \\ \frac{9}{2} & \frac{53}{2} \end{bmatrix} \end{aligned} \tag{37}$$

From equation 37 we can say that

$$\begin{aligned} &\text{Rank}\left(X_3\left(\tfrac{1}{2}\right)\right) = 2 \\ &\implies \text{Rank}\left(X_3\left(\tfrac{1}{2}\right)\right) > 1 \\ &\implies \text{Rank}\left(X_3\left(\tfrac{1}{2}\right)\right) > \tfrac{1}{2} + \tfrac{1}{2} \\ &\implies \text{Rank}\left(\tfrac{1}{2} X_1 + \tfrac{1}{2} X_2\right) > \tfrac{1}{2}\,\text{Rank}(X_1) + \tfrac{1}{2}\,\text{Rank}(X_2) \\ &\implies \text{obj}\left(\tfrac{1}{2} X_1 + \tfrac{1}{2} X_2\right) > \tfrac{1}{2}\,\text{obj}(X_1) + \tfrac{1}{2}\,\text{obj}(X_2). \end{aligned} \tag{38}$$

Hence, $\text{obj}(X)$ is not a convex function.

Hence the problem is not convex.

b) Nuclear Norm Minimization

We define function $f(X)$ as follows:

$$f(X) = \|X\|_* = \sup_{\|Y\|_2 \le 1} \langle X, Y \rangle \tag{39}$$

where $X \in \mathbb{R}^{p \times n}$ and $\|\cdot\|_2$ is the spectral norm. $\langle X, Y \rangle$ is given by

$$\langle X, Y \rangle := \text{Trace}\left(X^T Y\right) \tag{40}$$

To prove : $f(X)$ is convex.

For this, we consider two points, $X_1, X_2 \in \mathbb{R}^{p \times n}$. We assume that X_1 and X_2 are such that

$$\|X_1\|_* = N_{X_1} \tag{41}$$
$$\|X_2\|_* = N_{X_2} \tag{42}$$

Now, we discuss a few points regarding the nuclear norm.

The nuclear norm function is given by

$$\|X\|_* = \sup_{\|Y\|_2 \le 1} \langle X, Y \rangle \tag{43}$$

where $X \in \mathbb{R}^{p \times n}$ and $Y \in \mathbb{R}^{p \times n}$.
In equation (43), the supremum is attained as:

1. For a given $X \in \mathbb{R}^{p\times n}$, the inner product $\langle X, Y\rangle$ is a linear function in Y. Hence, for a given X, $\langle X, Y\rangle$ is a continuous function of Y.
2. The set of allowed matrices Y, which is given by set S:
$$S = \left\{Y \in \mathbb{R}^{p\times n} : \|Y\|_2 \leq 1\right\} \tag{44}$$
is a compact set.
3. The **Extreme Value Theorem** tells us that if K is a compact set and we have a function
$$f : K \to \mathbb{R}$$
which is continuous, then f is bounded on K and attains both its supremum and infimum.

Hence we have proved that the supremum is attained in the nuclear norm.

Coming back to X_1 and X_2, we defined in equation (43), we assume the following:

$$Y_1 = \arg\max_{\|Y\|_2\leq 1} \langle X_1, Y\rangle \tag{45}$$
$$Y_2 = \arg\max_{\|Y\|_2\leq 1} \langle X_2, Y\rangle \tag{46}$$

Using equations (41) and (45), we can say

$$\|X_1\|_* = N_{X_1} = \text{Trace}(X_1^T Y_1) \tag{47}$$

Similarly,

$$\|X_2\|_* = N_{X_2} = \text{Trace}(X_2^T Y_2) \tag{48}$$

We define $X_3(\lambda) \in \mathbb{R}^{p\times n}$ as the convex combination of X_1 and X_2:

$$X_3(\lambda) = \lambda X_1 + (1-\lambda)X_2 \qquad \text{for } \lambda \in [0,1] \tag{49}$$

The nuclear norm of $X_3(\lambda)$ is given by

$$\|X_3(\lambda)\|_* = \sup_{\|Y\|_2\leq 1} \langle X_3(\lambda), Y\rangle \tag{50}$$

$$\begin{aligned}\|X_3(\lambda)\|_* &= \sup_{\|Y\|_2\leq 1} \text{Trace}\left(X_3(\lambda)Y\right)\\ &= \sup_{\|Y\|_2\leq 1} \text{Trace}\left((\lambda X_1 + (1-\lambda)X_2)Y\right)\\ &= \sup_{\|Y\|_2\leq 1} \left[\lambda\text{Trace}(X_1Y) + (1-\lambda)\text{Trace}(X_2Y)\right]\end{aligned} \tag{51}$$

As $\text{Trace}(XY)$ is linear in X,

$$\|X_3(\lambda)\|_* \leq \lambda N_{X_1} + (1-\lambda)N_{X_2} \tag{52}$$
$$\|X_3(\lambda)\|_* \leq \lambda f(X_1) + (1-\lambda)f(X_2) \tag{53}$$

Hence, from equation (53), we have

$$f(X_3(\lambda)) \leq \lambda f(X_1) + (1-\lambda)f(X_2) \tag{54}$$

for all $X_1, X_2 \in \mathbb{R}^{p\times n}$ and $\lambda \in [0,1]$.

The optimization problem given by

$$\begin{aligned}\min_{X\in\mathbb{R}^{p\times n}} \quad & \|X\|_*\\ \text{subject to} \quad & AX = B\end{aligned} \tag{55}$$

is convex in nature as:

1. We proved that the objective function, $\|X\|_*$, is convex.
2. The constraint set
$$\{X \in \mathbb{R}^{p\times n} : AX = B\} \tag{56}$$
is a convex set (proved earlier).

c 1)

Consider a linear time-invariant (LTI) system which has an impulse response $X_{\text{LTI-system}}$ of length 5 ($X_{\text{LTI-system}} \in \mathbb{R}^5$).
We denote it as follows:

$$X_{\text{LTI-system}} = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \tag{57}$$

Now consider an input signal A_{input} of length 5 ($A_{\text{input}} \in \mathbb{R}^5$). A_{input} is given as follows:

$$A_{\text{input}} = \begin{bmatrix} u_0 \\ u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} \tag{58}$$

When A_{input} is passed as input into the LTI system with impulse response $X_{\text{LTI-system}}$ as given in equation (57), the output, $\text{B}_{\text{output}} \in \mathbb{R}^5$, is given as:

$$\text{B}_{\text{output}} = \begin{bmatrix} x_0u_0 \\ x_0u_1 + x_1u_0 \\ x_0u_2 + x_1u_1 + x_2u_0 \\ x_0u_3 + x_1u_2 + x_2u_1 + x_3u_0 \\ x_0u_4 + x_1u_3 + x_2u_2 + x_3u_1 + x_4u_0 \end{bmatrix} \tag{59}$$

B_{output} in equation (59) can also be expressed as:

$$\text{B}_{\text{output}} = \begin{bmatrix} u_0 & 0 & 0 & 0 & 0 \\ u_1 & u_0 & 0 & 0 & 0 \\ u_2 & u_1 & u_0 & 0 & 0 \\ u_3 & u_2 & u_1 & u_0 & 0 \\ u_4 & u_3 & u_2 & u_1 & u_0 \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = AX_{\text{LTI-system}} \tag{60}$$

Where $A \in \mathbb{R}^{5\times5}$ is a lower triangular Toeplitz matrix made using the input signal A_{input} (see equation (58)).

This shows that convolution is a linear operation on the impulse response of the LTI system.

We construct A (which is lower triangular Toeplitz) from A_{input} so that convolution is automatically performed as shown in equation (60), when we multiply A with $X_{\text{LTI-system}}$

Hence, matrix A has the special lower triangular Toeplitz structure here, to make discrete time convolution into a simple matrix multiplication operation.

In the question given here, we apply the signal $u \in \mathbb{R}^5$ given by

$$u = \begin{bmatrix} 1.0 \\ 0.8 \\ 0.2 \\ 0.5 \\ 0 \end{bmatrix} \tag{61}$$

to an LTI system for a total of 5 times.

We denote the impulse response of the LTI system for the i-th experiment as x^i ($x^i \in \mathbb{R}^5,\ i = 1, 2, 3, 4, 5$):

$$x^i = \begin{bmatrix} x_0^i \\ x_1^i \\ x_2^i \\ x_3^i \\ x_4^i \end{bmatrix} \tag{62}$$

The response of LTI system for the ith experiment is given by

$$\text{output} = Ax^i \tag{63}$$

where $A \in \mathbb{R}^{5\times5}$ is the Toeplitz matrix constructed from input u in equation (61) and is given by

$$A = \begin{bmatrix} u_0 & 0 & 0 & 0 & 0 \\ u_1 & u_0 & 0 & 0 & 0 \\ u_2 & u_1 & u_0 & 0 & 0 \\ u_3 & u_2 & u_1 & u_0 & 0 \\ u_4 & u_3 & u_2 & u_1 & u_0 \end{bmatrix} \tag{64}$$

Let $B \in \mathbb{R}^{5\times5}$ be the output of all 5 experiments.
Hence matrix B is given by

$$B = A\begin{bmatrix} x^1 & x^2 & x^3 & x^4 & x^5 \end{bmatrix} = AX \tag{65}$$

where $X \in \mathbb{R}^{5\times5}$ is the matrix whose jth column is x^j, $j = 1, 2, 3, 4, 5$.

In the question, we are given matrices A and B, and we wish to find matrix X.

It is often noticed that in such settings, the impulse responses are similar across experiments, i.e.,

$$\text{rank}(X) < 5 \tag{66}$$

Hence, to estimate matrix X, we need to solve the following optimization problem:

$$\begin{aligned} \min_{X\in\mathbb{R}^{5\times5}} \quad & \text{rank}(X) \\ \text{s.t. } & AX = B \end{aligned} \tag{67}$$

We cannot solve the optimization problem in equation (67) directly as $\text{rank}(X)$, as we showed previously, is not a convex function.

Instead, we formulate its convex relaxation, which we aim to solve.

Convex relaxation of equation (67)**:**

$$\begin{aligned} \min_{X\in\mathbb{R}^{5\times5}} \quad & \|X\|_* \\ \text{s.t. } & AX = B \end{aligned} \tag{68}$$

Why should the convex relaxation in equation (68) yield an X close to the X which solves equation (67)?

1. The nuclear norm $\|X\|_*$ is the convex envelope of $\text{rank}(X)$ on the spectral norm unit ball:
$$\left\{X \in \mathbb{R}^{m\times n} : \|X\|_2 \le 1\right\}$$
2. On the set
$$\left\{X \in \mathbb{R}^{m\times n} : \|X\|_2 \le \alpha\right\}$$
for some $\alpha > 0$, the convex envelope of $\text{rank}(X)$ is $\frac{1}{\alpha}\|X\|_*$ where $\alpha > 0$ and $\alpha \in \mathbb{R}$.
3. Here, our feasible set is given by
$$S = \{X \in \mathbb{R}^{5\times5} : AX = B\}$$
For all X belonging to S, there will always exist $\beta > 0$, $\beta \in \mathbb{R}$, such that $\|X\|_* \le \beta$ for all $X \in S$ (in fact, since $AX = B$ contains only one element (for the A and B given in this question), as it is feasible and matrix A is invertible, this bound is trivial).
In that case, $\beta^{-1}\|X\|_*$ will be the convex envelope of $\text{rank}(X)$ when we consider S as the domain.
4. Since the convex envelope of a function is the largest convex function lying below it on the domain, minimizing the convex envelope often yields an X that also minimizes the original function. In particular,
$$\arg\min_{X\in\mathbb{R}^{5\times5}} \text{rank}(X) \;\simeq\; \arg\min_{X\in\mathbb{R}^{5\times5}} \frac{1}{\beta}\|X\|_* \quad \text{subject to } AX = B, \tag{69}$$
where $\frac{1}{\beta}\|X\|_*$ is the convex envelope of $\text{rank}(X)$ when we take the constraint set as domain.

5. Solving the problem

$$\min_{X\in\mathbb{R}^{5\times 5}} \frac{1}{\beta}\|X\|_* \quad \text{s.t.} \quad AX = B, \qquad \beta > 0,\ \beta \in \mathbb{R}$$

is the same as solving

$$\min_{X\in\mathbb{R}^{5\times 5}} \|X\|_* \quad \text{s.t.} \quad AX = B$$

Hence, solving the convex relaxation of the problem in equation 68 yields an X close to the one which solves the problem defined in equation 67.

Moreover, in the paper [2], the authors explain how we can replace the problem in 67 by the one in 68 and convert it into a semi definite program so that it can be solved by available solvers.

$$\text{Optimizer} = \begin{bmatrix} 1 & -1 & 0 & 0 & 1 \\ 2 & 0 & 1 & -1 & 0 \\ 1 & 1 & 1 & -1 & -1 \\ 2 & -2 & 0 & 0 & 2 \\ -2 & 0 & -1 & 1 & 0 \end{bmatrix}$$

c 2)

Previously, we had to solve the problem defined in

$$AX = B \tag{70}$$

where $AX = B$ was feasible.

Now, we replace B by $B + N$, where $N \in \mathbb{R}^{5\times 5}$ is given by

$$N[i,j] \sim \mathcal{N}(0, 0.1), \qquad i = j \tag{71}$$

$$N[i,j] = 0, \qquad i \neq j \tag{72}$$

Hence, $AX = B + N$ may not be feasible anymore.

We must modify our formulation to:

(a) Obtain $X \in \mathbb{R}^{5\times 5}$ such that it minimizes $\text{rank}(X)$ and satisfies $AX = B + N$, when it is feasible.

(b) It should also be able to handle the case where $AX = B + N$ is not feasible.

Satisfying the points mentioned above, we define the new optimization problem to solve as follows:

$$\min_{X\in\mathbb{R}^{5\times 5}} \lambda \|AX - (B + N)\|_F^2 + \text{rank}(X) \tag{73}$$

where:

(a) $\|\cdot\|_F$ denotes the Frobenius norm of a matrix. For an $m \times n$ matrix $A = [a_{ij}]$, $\|A\|_F$ is given as

$$\|A\|_F = \sqrt{\sum_{i=1}^{m}\sum_{j=1}^{n}(a_{ij})^2} \tag{74}$$

(b) λ is a parameter we choose which controls the trade-off between least-squares fit to measured B and promoting low-rank.

We typically set λ of the order of 10 or so, as $AX = B + N$ must be satisfied when feasible.

We solve the convex relaxation of the modified formulation, which is given as

$$\min_{X \in \mathbb{R}^{5\times 5}} \lambda \|AX - (B+N)\|_F^2 + \|X\|_* \tag{75}$$

where $\|\cdot\|_*$ denotes the nuclear norm.

$$\text{Optimizer} = \begin{bmatrix} 0.98 & -1.02 & -0.02 & -0.02 & 1.01 \\ 1.94 & -0.01 & 1.01 & -0.97 & -0.03 \\ 1.06 & 0.98 & 1.07 & -1.03 & -0.92 \\ 1.90 & -1.96 & -0.03 & -0.04 & 1.89 \\ -1.86 & -0.01 & -0.96 & 0.99 & -0.06 \end{bmatrix}$$

Question 3

Reference used for this question : [3]

In this question, we have an undirected graph

$$G = (V, E) \tag{76}$$

where V is the set of vertices and E is the set of edges.

We are given

$$|V| = n \tag{77}$$

We have a set of K colours and we must assign to each $v \in V$ a colour in

$$\{1, 2, 3, \ldots, K\} \tag{78}$$

The assignment of colors must be done so that no two adjacent vertices (vertices connected by an edge) receive the same colour.

Here the number of colours, K, is a variable and the smallest value of K for which a legal colouring, which satisfies the constraints, is possible is called the *chromatic number* (denoted by $\chi(G)$).

3a) Feasibility via a 0–1 formulation

We now move to vectorising the graph colouring problem.

A coloured graph, where we have colours $1, 2, 3, \ldots, K$ and N vertices, can be represented by a vector of length NK, where the individual entries of the vector are given as

$$x_{vk} \qquad \text{where } v \in V,\ k \in \{1, 2, \ldots, K\} \tag{79}$$

with the binary constraint $x_{vk} \in \{0, 1\}$.

For the graph being represented by the vector to be a valid graph, we need the vector with x_{vk} as its elements to satisfy the following:

(a)

$$x_{vk} \in \{0, 1\} \tag{80}$$

where

$$x_{vk} = \begin{cases} 1 & \text{if vertex } v \text{ is assigned colour } k, \\ 0 & \text{otherwise.} \end{cases}$$

(b) Each vertex is coloured with exactly one colour.

$$\sum_{k=1}^{K} x_{vk} = 1 \qquad (\forall v \in V) \tag{81}$$

(c) No two adjacent vertices have the same colour.

$$x_{uk} + x_{vk} \leq 1 \qquad \forall (u, v) \in E,\ \forall k \tag{82}$$

3 a) 1)

For a given graph G and K such that $K \geq \chi(G)$, let *allowed_colouring* denote the set of vectors which satisfy the above mentioned constraints.

We now need to determine whether the set *allowed_colouring* is convex or not.

The set *allowed_colouring* is **not** convex as:

(a) For any vector belonging to the set, each entry of the vector $x_{vk} \in \{0, 1\}$.

(b) However, we can form convex combinations of two vectors belonging to this set such that the entries of the convex combination $\in (0, 1)$ and $\notin \{0, 1\}$.

(c) Hence, it is possible to construct convex combinations of the vectors from the set such that the convex combination $\notin$ *allowed_colouring*.

Example:

Consider the following graph G with two nodes and one edge:

$$1 \longleftrightarrow 2$$

$$G = (V, E)$$

where: V = {1, 2} and E = {(1,2)}

Let $K = 2$.

The general form of a vector representing a coloured graph is given as

$$\text{general_coloured_graph} = \begin{pmatrix} x_{11} \\ x_{12} \\ x_{21} \\ x_{22} \end{pmatrix}$$

where $x_{vk} \in \{0, 1\}$ $\qquad (v \in V,\ k \in \{1, 2\})$

The set *allowed_colouring* is given by

$$\textit{allowed_colouring} = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix} \right\} = \{x_1, x_2\} \quad \text{(suppose)}$$

A convex combination of x_1 and x_2 (allowed_colouring) is given by

$$x_3(\lambda) = \lambda x_1 + (1 - \lambda) x_2 \qquad \text{where } \lambda \in [0, 1]$$

Taking $\lambda = \frac{1}{2}$,

$$x_3\left(\frac{1}{2}\right) = \frac{1}{2}x_1 + \frac{1}{2}x_2 = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} \notin \text{ allowed_colouring}$$

Hence the set *allowed_colouring* is not convex.

3 a) 2)

Now the binary constraint that

$$x_{vk} \in \{0,1\}, \quad (v \in V,\ k \in \{1,2,\ldots,K\})$$

has been dropped and is instead replaced by

$$0 \leq x_{vk} \leq 1$$

Hence, our definition for the set *allowed_colouring* now changes. The new definition for the set *allowed_colouring* after taking this relaxation into account is given by:

$$\begin{aligned} \textit{allowed_colouring} = \Bigg\{ x \in \mathbb{R}^{KN} \ &: x_{vk} \in [0,1], \\ &\sum_{k=1}^{K} x_{vk} = 1 \quad (\forall v \in V), \\ &x_{uk} + x_{vk} \leq 1 \quad (\forall (u,v) \in E, \forall k) \Bigg\} \end{aligned} \tag{83}$$

where $|V| = N$.

The vector $x \in \mathbb{R}^{KN}$ in equation 83 has the form:

$$x = \begin{pmatrix} x_{11} \\ x_{12} \\ \vdots \\ x_{1K} \\ x_{21} \\ \vdots \\ x_{2K} \\ \vdots \\ x_{N1} \\ x_{N2} \\ \vdots \\ x_{NK} \end{pmatrix}$$

Now we wish to tell if the new set, allowed_colouring defined in equation (83), is convex or not.
For this, consider two points

$$x^1,\ x^2 \in \text{allowed_colouring}$$

Assume x^1 and x^2 have the following form:

$$x^1 = \begin{pmatrix} x^1_{11} \\ x^1_{12} \\ \vdots \\ x^1_{1K} \\ x^1_{21} \\ \vdots \\ x^1_{2K} \\ \vdots \\ x^1_{N1} \\ x^1_{N2} \\ \vdots \\ x^1_{NK} \end{pmatrix}, \quad x^2 = \begin{pmatrix} x^2_{11} \\ x^2_{12} \\ \vdots \\ x^2_{1K} \\ x^2_{21} \\ \vdots \\ x^2_{2K} \\ \vdots \\ x^2_{N1} \\ x^2_{N2} \\ \vdots \\ x^2_{NK} \end{pmatrix}$$

Since $x^1, x^2 \in$ allowed_colouring defined in equation (83), their components satisfy the following constraints:

$$x^\alpha_{vk} \in [0, 1] \tag{84}$$

where $\alpha \in \{1, 2\}$, $v \in V$, $k \in \{1, 2, \ldots, K\}$.

$$\sum_{k=1}^{K} x^\alpha_{vk} = 1 \qquad (\forall v \in V) \tag{85}$$

$\alpha \in \{1, 2\}$

$$x^\alpha_{uk} + x^\alpha_{vk} \leq 1 \qquad (\forall (u, v) \in E, \forall k) \tag{86}$$

$\alpha \in \{1, 2\}$

Consider a convex combination of x^1 and x^2. Let the convex combination be $x^3(\lambda)$. $x^3(\lambda)$ is given by

$$x^3(\lambda) = \lambda x^1 + (1 - \lambda) x^2 \tag{87}$$

where $\lambda \in [0, 1]$.

Now, we wish to determine whether $x^3(\lambda) \in$ allowed_colouring or not.

1) Checking if $x^3_{vk} \in [0, 1]$:

$$x^3_{vk} = \lambda x^1_{vk} + (1 - \lambda) x^2_{vk} \tag{88}$$

for $v \in V$, $k \in \{1, 2, \ldots, K\}$.

Since equation (88) is satisfied and $\lambda \in [0, 1]$, we can say that

$$x^3_{vk} \in [0, 1], \qquad \forall v \in V, k \in \{1, 2, \ldots, K\} \tag{89}$$

Therefore, this condition is satisfied.

2) Checking if

$$\sum_{k=1}^{K} x_{vk}^3 = 1 \qquad \forall v \in V \tag{90}$$

As mentioned earlier,

$$x_{vk}^3 = \lambda x_{vk}^1 + (1-\lambda) x_{vk}^2 \tag{91}$$

for $v \in V$ and $k \in \{1, 2, \ldots, K\}$.
Hence, we have

$$\begin{aligned}
\sum_{k=1}^{K} x_{vk}^3 &= \sum_{k=1}^{K} \left(\lambda x_{vk}^1 + (1-\lambda) x_{vk}^2\right) \\
&= \lambda \sum_{k=1}^{K} x_{vk}^1 + (1-\lambda) \sum_{k=1}^{K} x_{vk}^2 \\
&= \lambda(1) + (1-\lambda)(1) \\
&= 1
\end{aligned} \tag{92}$$

Hence, this condition is satisfied.

3) Checking if

$$x_{uk}^3 + x_{vk}^3 \leq 1 \qquad \forall (u,v) \in E,\ \forall k \tag{93}$$

We have

$$x_{vk}^3 = \lambda x_{vk}^1 + (1-\lambda) x_{vk}^2 \tag{94}$$

for $v \in V$, $k \in \{1, 2, \ldots, K\}$.
Using equation (94):

$$\begin{aligned}
x_{uk}^3 + x_{vk}^3 &= \lambda x_{uk}^1 + (1-\lambda) x_{uk}^2 + \lambda x_{vk}^1 + (1-\lambda) x_{vk}^2 \\
&= \lambda(x_{uk}^1 + x_{vk}^1) + (1-\lambda)(x_{uk}^2 + x_{vk}^2)
\end{aligned} \tag{95}$$

Using the original constraints,

$$x_{uk}^1 + x_{vk}^1 \leq 1, \quad x_{uk}^2 + x_{vk}^2 \leq 1$$

we have

$$x_{uk}^3 + x_{vk}^3 \leq \lambda(1) + (1-\lambda)(1) = 1$$

Thus,

$$x_{uk}^3 + x_{vk}^3 \leq 1 \qquad \forall (u,v) \in E,\ \forall k \tag{96}$$

Therefore, the third condition has also been satisfied.

Since all 3 conditions have been satisfied,

$$x^3(\lambda) \in \text{allowed_colouring} \qquad \forall \lambda \in [0,1] \tag{97}$$

Since our choice of x^1 and $x^2 \in$ allowed_colouring was arbitrary, the new set allowed_colouring is convex.

3 b) 1)

We now associate each vertex $v \in V$ with a unit vector u_v.

We construct the Gram matrix as follows:

$$G_{uv} = u_u^\top u_v \tag{98}$$

for all $u, v \in V$.

1) For a given K ($K \geq 2$), we say that a set of vectors

$$\mathcal{U} = \{u_1, u_2, \ldots, u_N\} \tag{99}$$

is feasible if the Gram matrix constructed using this set of vectors (according to equation (98)) satisfies the following:

(a) $G \in S^n$,

(b) $G \succeq 0$,

(c) $G_{vv} = 1 \qquad \forall v \in V$,

(d) $G_{uv} \leq -\frac{1}{K-1} \qquad \forall (u, v) \in E$

If for a given K, finding such a set of vectors is possible, then the K-colouring of the graph is possible.

If for a given K, such a set of vectors satisfying the above constraints can never be found, then the K-colouring of the graph is not possible.

3 b) 2)

The optimization problem minimizing the edge-wise inner product bound ρ is given by:

$$\min_{G,\,\rho} \ \rho \quad \text{s.t.} \quad G \succeq 0, \quad G_{vv} = 1 \ (\forall v \in V), \quad G_{uv} \leq \rho \ (\forall (u, v) \in E) \tag{100}$$

The optimization problem above is equivalent to the feasibility problem described in the previous sub-part as:

(a) The feasibility problem in the previous sub-part aims to tell if K-coloring of a graph is possible or not, by checking if it is possible to form a set of unit vectors of size N ($|V| = N$), such that the edge-wise inner product is upper bounded by $-\frac{1}{K-1}$.

(b) The optimization problem in equation (100) aims to tell the lowest possible upper bound for the edge-wise inner product ρ when we construct a set of N unit vectors.

Due to the equivalence of the two problems, we can use the solution of the optimization problem in equation (100) to determine the feasibility of the K-coloring of the graph.

Let

$$\rho^* = \text{solution of the optimization problem} \tag{101}$$

The K-coloring of the graph would be possible iff:

$$\rho^* \leq -\frac{1}{K-1} \tag{102}$$

3 b) 3)

Let

$$\rho^* = \text{solution to the optimization problem} \tag{103}$$

From the previous two parts of the question, we can say that the K-colouring of the graph is possible if and only if

$$\rho^* \leq -\frac{1}{K-1} \tag{104}$$

which implies

$$-\rho^* \geq \frac{1}{K-1} \tag{105}$$

and hence

$$-\frac{1}{\rho^*} \leq K-1 \tag{106}$$

which can be rewritten as

$$1-\frac{1}{\rho^*} \leq K \tag{107}$$

We have:

Chromatic number of G (denoted $\chi(G)$) is the smallest value of K for which K-colouring is possible.
Using equation (107), we can say that

$$\chi(G) = \text{smallest integer greater than } \left(1+\frac{1}{-\rho^*}\right) \tag{108}$$

Therefore,

$$\chi(G) = \left\lceil 1+\frac{1}{-\rho^*} \right\rceil + 1 \tag{109}$$

where $\lceil \cdot \rceil$ denotes the greatest integer (floor) function.

3 c)

One of the practical applications of the graph coloring problem is in the preparation of exam timetables by a university.
Here:

(a) Colours are replaced by time slots given to the exam. The nodes of the graph are exams.
(b) If there is a common student between two exams, a connection (edge) is made between the corresponding exam nodes.
(c) Hence, if there is a student who is to attempt multiple exams, the corresponding exam nodes need to be given different time slots (exam nodes connected by edges need to be coloured differently).

Our goal is to minimize the number of time slots required to conduct the exams.
Let `Exams` be the set containing all the exams to be conducted.
Let

$$|\texttt{Exams}| = N$$

(N number of exams need to be conducted)
Let `Edges` be the set containing the different edges between exam nodes in our graph.
Hence the graph is given by

$$\text{Graph} = G(\texttt{Exams}, \texttt{Edges})$$

We need to find the minimum number of time slots required to conduct all exams, or equivalently, the minimum number of colors to color the graph, such that no two adjacent exams (connected by an edge) occur in the same time slot.

For this, we need to solve the following optimization problem:

$$\min_{G,\,\rho} \ \rho \quad \text{s.t.} \quad G \in S^N, \tag{110}$$

$$G \succeq 0, \tag{111}$$

$$G_{vv} = 1 \qquad \text{where } v \in \text{Exams}, \tag{112}$$

$$G_{uv} \leq \rho \qquad \forall (u,v) \in \text{Edges} \tag{113}$$

where $u \in$ Exams, $v \in$ Exams.

Let ρ^* be the solution to the optimization problem above.

Then, we can hold the exams in K slots, without any exam clashing for students, if

$$\rho^* \leq -\frac{1}{K-1} \tag{114}$$

which implies

$$1 - \frac{1}{\rho^*} \leq K \tag{115}$$

Since we want the number of slots required to conduct exams to be minimized, we set

$$\text{minimum number of slots needed} = \left\lceil 1 + \frac{1}{-\rho^*} \right\rceil \tag{116}$$

where $\lceil \cdot \rceil$ denotes the greatest integer (ceiling) function.

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